

Unit 1 Review Sheet

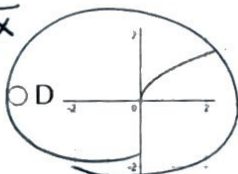
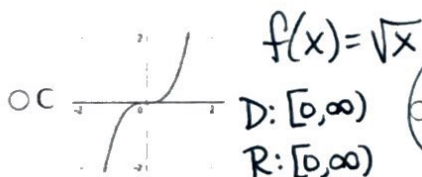
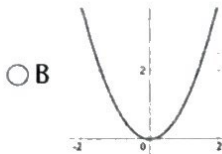
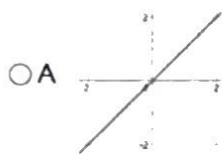
Key

What do I need to know?

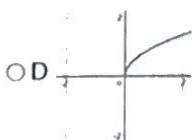
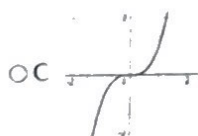
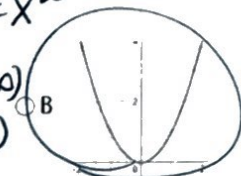
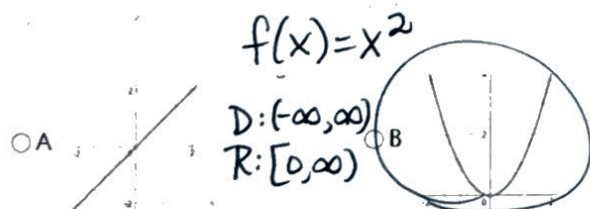
1. Given a graph, identify the parent function.
2. Given the equation of a function, know what the graph looks like
3. Given an equation, be able to identify the domain, range, and x and y intercepts (if present).
4. State the symmetry of the graph of the function. Is it even (y-axis symmetry) or odd (origin symmetry)?
5. Given a real-world scenario, be able to state independent and dependent variables
6. Given a real-world scenario, be able to state a reasonable domain and range.

REVIEW PROBLEMS:

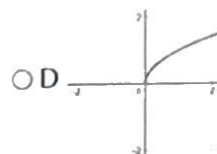
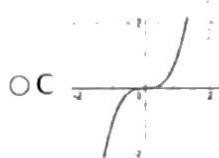
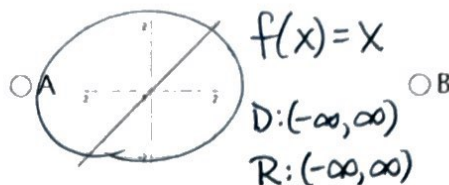
1. Which one of the following is the square root parent function?



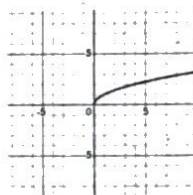
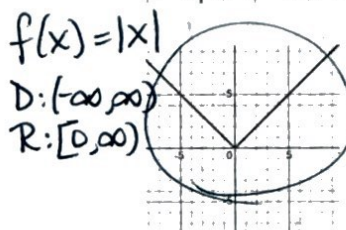
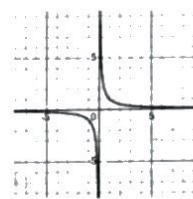
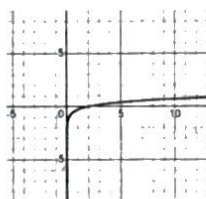
2. Which one of the following is the quadratic parent function?



3. Which of the following is the linear parent function?



4. Which of the following is the absolute value parent function?



5. List the equations for all of the parent functions we are focusing on in this unit that have a domain of $x \geq 0$

$$f(x) = \sqrt{x}$$

6. List the equations for all of the parent functions we are focusing on in this unit that have a range $y \geq 0$

$$f(x) = \sqrt{x}$$

$$f(x) = x^2$$

$$f(x) = |x|$$

7. Which parent functions we are focusing on in this unit have symmetry about the origin? (Odd)

linear

8. Which parent functions we are focusing on in this unit have symmetry about the y-axis? (even)

quadratic & absolute value

9. A phone company charges a base fee of \$30 per month and an additional \$0.10 for each text message sent. Let the total monthly cost be C in dollars, and the number of text messages sent be x .

- a. Identify the independent and dependent variables, and explain why.

ind. (x) is the number of text messages and the dep. (C) is the cost, because it depends on how many text messages.

- b. State a reasonable domain and range for this situation

$D: x \geq 0$ it is reasonable that you can have no texts or more.

$R: C \geq 30$ your bill is 30 if no texts, but will only increase if there are texts.

10. The height h of a sunflower (in centimeters) grows over time t (in days) after it is planted. Suppose the sunflower grows at a constant rate of 2 cm per day for 50 days, then its growth slows to 0.5 cm per day for the next 30 days.

- a. Identify the independent and dependent variables.

the ind. variable is number of days
the dep. variable is the growth in cm.

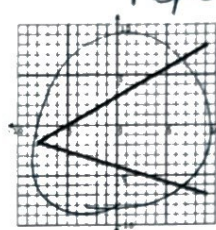
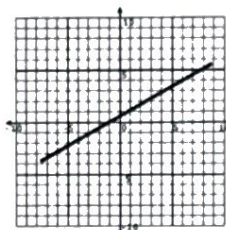
- b. State a reasonable domain and range for this situation.

$D: [0, 80]$ → no days, then 50, then 30

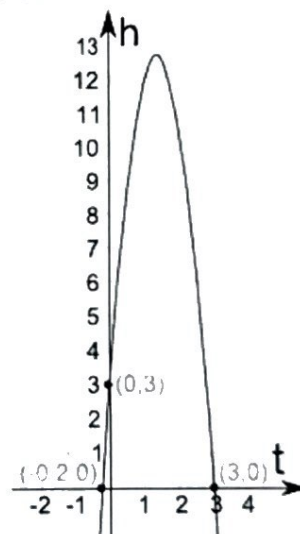
$R: [0, 115]$ → 2 cm/day for 50 = 100, then .5 cm/day for 30 = 15

11. Which one of these is not a function? Why?

the x-values repeat



12. Julian, on the edge of sanity, launches his backpack up into the air from some stairs, which are 3 feet high. The graph below models the path of his backpack as it travels through the panhandle sky. Analyze how long it takes his backpack to hit the ground.



- a. What does the domain represent?

the time in seconds

- b. What does the range represent?

the height of the backpack

- c. What is a reasonable domain for the scenario? $D: [0, 3]$

- d. What is a reasonable range for the scenario? $R: [0, 12.5]$

- e. What are the intercepts and what do they represent in this situation?

x-int: (3, 0) represents that at 3 seconds the backpack is on the ground. y-int: (0, 3) represents the initial height of the backpack before it is thrown.

13.

Determine the following for the function:

Domain(inequality): $\mathbb{R}$

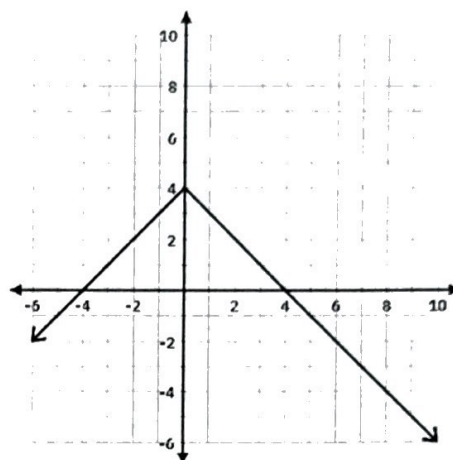
Range (inequality): $y \leq 4$

Max or Min: Max at (0,4)

x-intercept: (-4,0) (4,0)

y-intercept: (0,4)

Parent Function Family : absolute value
 $f(x) = |x|$



14.

Determine the following for the function:

Domain(interval): $(-\infty, \infty)$

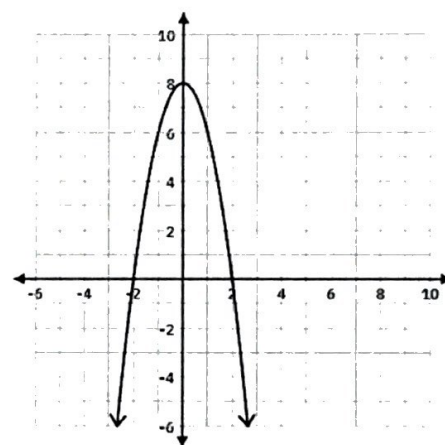
Range (interval): $(-\infty, 8]$

Max or Min: max at (0,8)

x-intercept: (-2,0) (2,0)

y-intercept: (0,8)

Parent Function Family : quadratic
 $f(x) = x^2$



15.

Determine the following for the function:

Domain(set): $\{x \mid x \geq 0; x \in \mathbb{R}\}$

Range (set): $\{y \mid y \geq 5; y \in \mathbb{R}\}$

Max or Min: min at (0,5)

x-intercept: none

y-intercept: (0,5)

Parent Function Family : Square root
 $f(x) = \sqrt{x}$

